

NLU Assignment 1

Sports vs Politics Text Classification

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1 Introduction

Text classification is an important task in Natural Language Processing (NLP) where textual data is automatically categorized into predefined classes. In this assignment, we implemented a classifier that distinguishes between **Sports** and **Politics** related text.

Three machine learning algorithms were compared:

- Naive Bayes
- Support Vector Machine (SVM)
- Logistic Regression

TF-IDF vectorization was used for feature extraction and model performance was evaluated using accuracy and classification metrics.

2 Dataset

A synthetic dataset was created inside the program consisting of short sentences representing two categories.

Sports examples include matches, players, tournaments and performance. **Politics** examples include government, elections, policies and leadership.

- Total samples: 40
- Sports: 20
- Politics: 20

The dataset is balanced to prevent classification bias.

3 Preprocessing

The following preprocessing steps were applied automatically:

- Lowercase conversion
- Stopword removal
- Tokenization

These steps reduce noise and improve classification accuracy.

4 Feature Extraction

TF-IDF converts text into numerical vectors.

$$TFIDF = TF \times IDF$$

Words unique to a category receive higher weight while common words receive lower importance.

5 Algorithms

5.1 Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes Theorem:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

The classifier assumes independence between words. Even though this assumption is unrealistic, it works well because certain words strongly indicate categories. For example, words like *goal*, *match*, *player* suggest sports while *government*, *election*, *policy* suggest politics.

Advantages:

- Fast training
- Works well on small datasets
- Effective for text classification

Limitations:

- Cannot understand context
- Assumes independence between words

5.2 Support Vector Machine

SVM separates classes using an optimal hyperplane with maximum margin. In text classification, TF-IDF generates high dimensional sparse vectors, which SVM handles efficiently.

Advantages:

- High accuracy
- Effective for sparse data
- Robust decision boundary

Limitations:

- Higher computational cost

5.3 Logistic Regression

Logistic Regression predicts probability using sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

It learns weights for words and classifies documents based on probability threshold.

Advantages:

- Interpretable
- Good baseline classifier

Limitations:

- Needs larger dataset for better performance

6 Training Setup

Dataset split:

- 75% training data
- 25% testing data

7 Results

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Loading dataset...
Total samples: 40

---- Model Comparison ----

Naive Bayes Accuracy: 90.00%
precision    recall  f1-score   support

   Politics      0.80      1.00      0.89         4
    Sports      1.00      0.83      0.91         6

 accuracy      0.90
macro avg      0.90      0.92      0.90         10
weighted avg   0.92      0.90      0.90         10

```

(a) Naive Bayes

```

Support Vector Machine Accuracy: 90.00%
precision    recall  f1-score   support

   Politics      0.80      1.00      0.89         4
    Sports      1.00      0.83      0.91         6

 accuracy      0.90
macro avg      0.90      0.92      0.90         10
weighted avg   0.92      0.90      0.90         10

```

(b) Support Vector Machine

```

Logistic Regression Accuracy: 80.00%
precision    recall  f1-score   support

 accuracy      0.80
macro avg      0.83      0.83      0.80         10
weighted avg   0.87      0.80      0.80         10

```

(c) Logistic Regression

```

Final Accuracy Comparison
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Naive Bayes           : 90.00%
Support Vector Machine : 90.00%
Logistic Regression   : 80.00%
PS D:\NluAssignment>

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(d) Accuracy Comparison

Figure 1: Model Outputs and Accuracy Comparison

8 Accuracy Comparison (Experimental Results)

Model	Accuracy
Naive Bayes	90%
Support Vector Machine	90%
Logistic Regression	80%

9 Discussion

From the obtained results, both Naive Bayes and Support Vector Machine achieved an accuracy of 90% while Logistic Regression achieved 80%. The classification report shows that the Politics class had perfect recall whereas the Sports class had higher precision. This indicates the model correctly identified most political samples but occasionally confused some sports sentences.

SVM performed similarly to Naive Bayes because TF-IDF produces high dimensional sparse features which both algorithms handle efficiently. Logistic Regression performed lower due to limited training samples and less optimized decision boundary.

10 Limitations

- Small dataset

- Short sentences only
- No contextual understanding

11 Future Work

Future improvements include using larger real-world datasets, applying deep learning models like LSTM and BERT, and incorporating n-gram features for better semantic understanding.

12 Conclusion

We implemented a sports vs politics classifier using TF-IDF features and three machine learning algorithms. SVM and Naive Bayes provided the best performance. This demonstrates traditional machine learning techniques remain effective for text classification tasks.